

Technical Document #937: Cybersecurity - Comprehensive Overview

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Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Intrusion detection systems (IDS) monitor network traffic for suspicious activity. They alert administrators of potential security breaches.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Intrusion detection systems (IDS) monitor network traffic for suspicious activity.
- Security information and event management (SIEM) systems collect and analyze security logs.
- Encryption converts plaintext into ciphertext to protect data confidentiality.